

MISFIT: A Self-Supervised Pretraining Toolkit for 3D Medical Imaging

A Preliminary Test Across Three BraTS 2026 Segmentation Tasks

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Introduction

Reported segmentation gains over nnU-Net are not always reproducible. We address this with **MIST**, our modular segmentation framework, which placed **3rd** in the 2024 and 2025 BraTS challenges. For BraTS 2026 we enter three tasks — brain metastases, pediatric tumors, and cross-population generalizability — each testing a different kind of generalization.

This report asks two questions

- 01** System-level: does a **MISFIT**-pretrained SwinUNETR-V2 pipeline beat our established pocket nnU-Net baseline?
- 02** Isolated effect: holding architecture, patch size, and budget fixed, does MAE pretraining itself help over random initialization?

Methods

Table 1 · Dataset sizes and regions evaluated per task				
TASK	COHORT	TRAIN	HELD-OUT	REGIONS
1	Brain metastases	1,196	179	NETC, SNFH, ET, RC
2	Pediatric tumors	294	91	ET, NET, CC, ED
3	BraTS-GoAT	1,351	450	ET, TC, WT

Self-supervised pretraining via MISFIT

MISFIT pretrains 3D medical-imaging encoders with masked autoencoding (MAE), independent of MIST. We pooled every training-split MRI volume across all three tasks into **16,315** single-channel volumes, masked 75% of each crop, and trained a SwinUNETR-base encoder for 200 epochs to reconstruct it. That encoder was then transferred into MIST’s SwinUNETR-base segmentation architecture.

Three network arms

- Pocket nnU-Net (baseline)** — our returning conv baseline, 128³ patch, random init.
- SwinUNETR-V2, MISFIT-pretrained** — 96³ patch, encoder from MISFIT weights.
- SwinUNETR-V2, from scratch** — identical, but randomly initialized — the matched control.

The two SwinUNETR arms are matched on everything but initialization, isolating the pretraining effect. The baseline differs in architecture, patch size, *and* budget too, so it tests the full pipeline instead.

Training & evaluation

5-fold cross-validation; the baseline trained up to 5,000 epochs/fold (mean 3,107–4,252 actual), both SwinUNETR arms exactly 1,000. Arms are compared via BraTS-style average rank (MIST’s mist_rank) and a two-sided Wilcoxon signed-rank test ($\alpha = 0.05$).

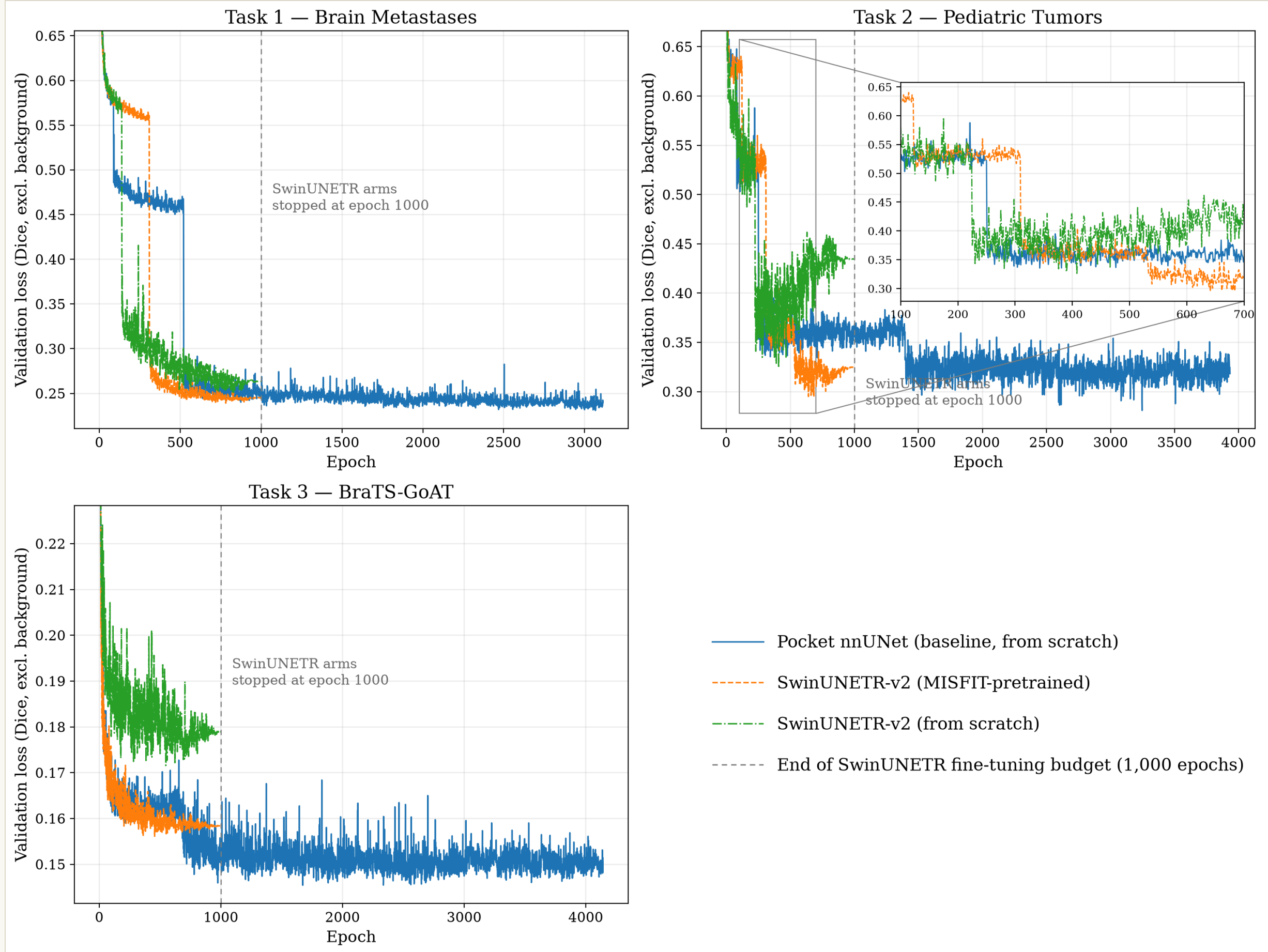


Figure 1. Validation loss across training. The MISFIT-pretrained arm reaches within 3.5–7.3% of the baseline’s converged floor inside its 1,000-epoch budget — far fewer epochs than the baseline needs. The from-scratch SwinUNETR fails to converge at all on Task 2.

Discussion & Conclusion

MAE pretraining genuinely helps once architecture, patch size, and budget are held fixed — but held-out validation does not support the pooled MISFIT pipeline beating our established baseline.

- Task 1’s reversal is explainable.** Its lesion-wise scoring amplifies a small boundary deficit that per-voxel CV Dice hides; 1,000 epochs likely wasn’t enough to refine it.
- Pretraining helps most where labels are scarce.** Task 2 (294 cases) sees the largest gain (ET Dice 0.518 → 0.670) and is the only task where the from-scratch model fails to converge.
- Final submission: pocket nnU-Net for all three tasks** — required by data-usage rules, and independently supported by the held-out results.
- This is a preliminary MAE schedule** — 200 epochs against the 800–1,600 convention. Longer pretraining may close the gap.

Results

Table 2 · BraTS-style average rank (lower is better), two-sided Wilcoxon p-values						
TASK	5-fold cross-validation			Held-out validation		
	BASE	MISFIT	P	BASE	MISFIT	P
1	1.503	1.497	0.62	1.427	1.573	1.5×10⁻⁴
2	1.528	1.472	0.026	1.503	1.497	0.83
3	1.518	1.482	0.0072	1.513	1.487	0.32
Task	MISFIT	Scratch	p	MISFIT	Scratch	p
1	1.439	1.561	2.1×10⁻⁵	1.436	1.564	0.0024
2	1.457	1.543	0.035	1.440	1.560	0.0053
3	1.416	1.584	1.9×10⁻¹³	1.424	1.576	7.4×10⁻⁵

Against the **matched control**, MISFIT pretraining wins on all three tasks under both protocols — the isolated effect is real. Against the **established baseline**, the picture reverses: CV favors the pretrained pipeline on Tasks 2–3, but that vanishes on held-out validation, where the baseline pulls significantly ahead on Task 1 ($p = 1.5\times10^{-4}$).

Why CV doesn’t replicate — the MAE encoder saw pooled training-split images label-free, inflating CV scores for Tasks 2–3. Held-out scoring excludes the pretraining pool entirely, which is why we treat it as decisive.

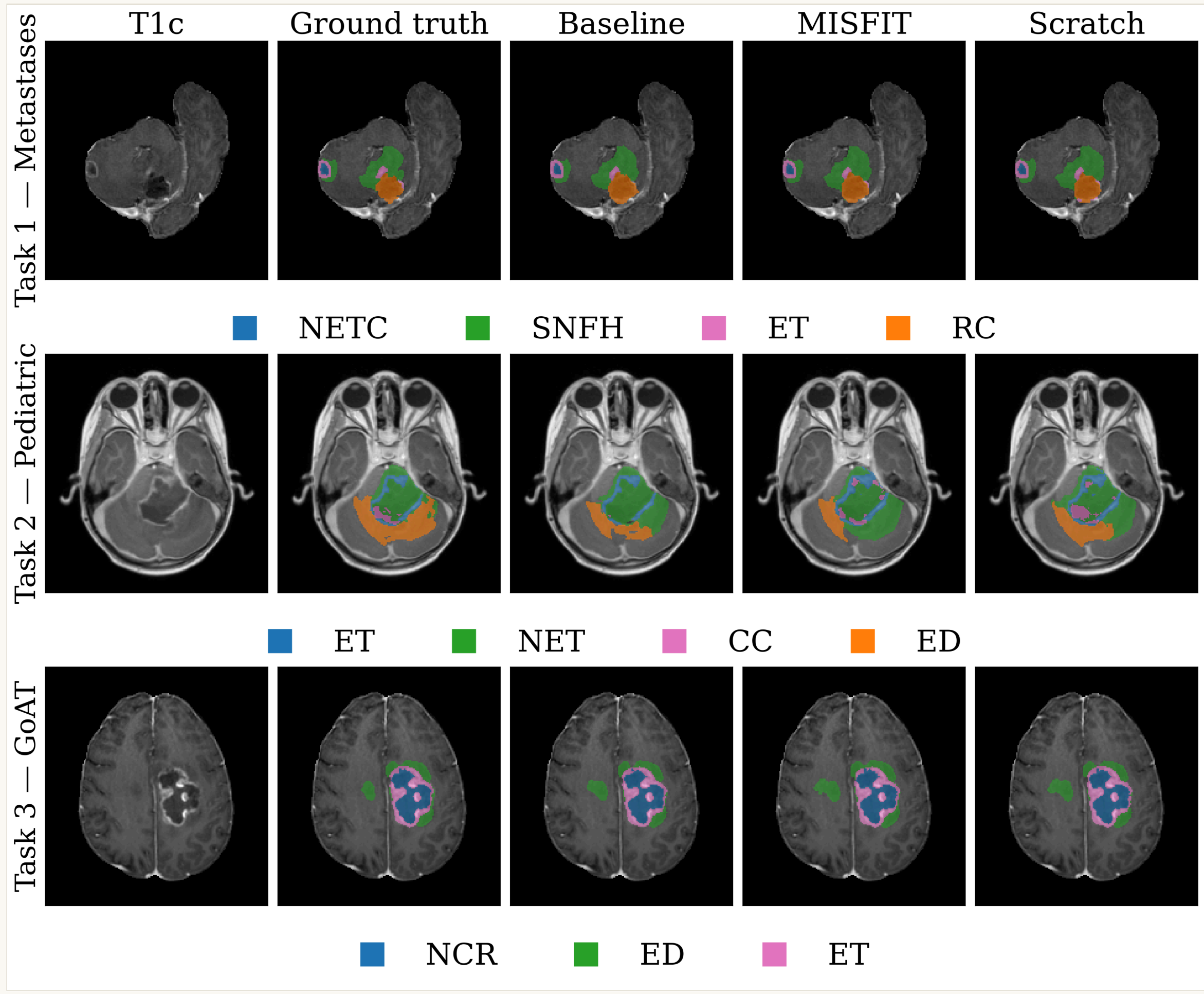


Figure 2. One representative case per task: T1c, ground truth, and all three arms. All agree closely on Tasks 1 and 3; on Task 2 the baseline recovers almost none of the cystic component (pink).